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Dose Accumulation in Adaptive Radiotherapy

a bridge between
what we treat
and what we judge

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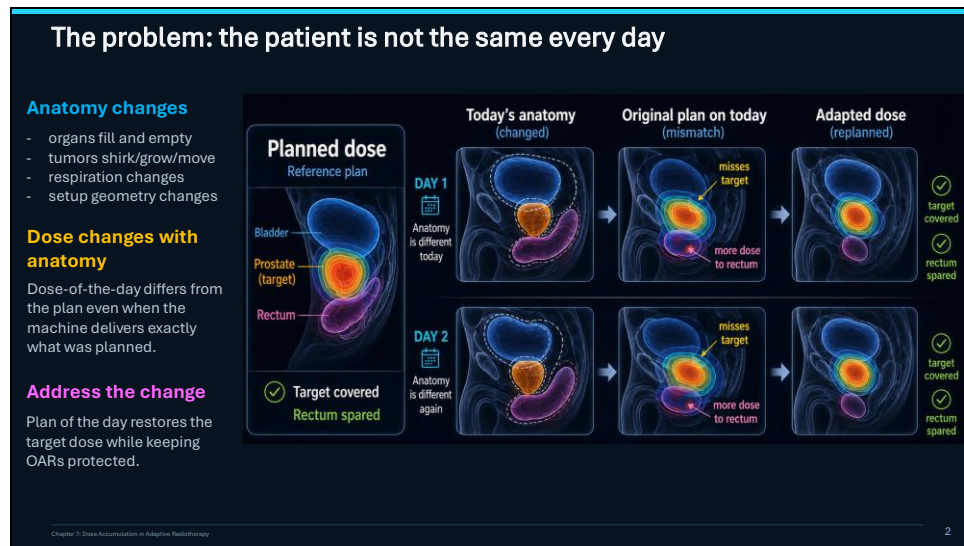
AMERICAN ASSOCIATION
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Treat fractions. Judge total.

Chapter 7: Dose Accumulation in Adaptive Radiotherapy

My name is Mihaela Rosu-Bubulac, from Virginia Commonwealth University and today I will talk about dose accumulation in adaptive radiotherapy as a bridge between what we treat and what we judge.

... because in radiotherapy we treat fractions, but clinically we judge the total treatment.



So let's start with the basic problem that takes us to the concept of dose accumulation: the patient is not the same every day.

As we all know, the treatment is planned on the patient anatomy from the simulation, such that the target is covered, and the OARs are spared.

But -

... the patient's anatomy changes over time:

Day 1 – anatomy is different, Day 2 – anatomy is different again, etc, because:

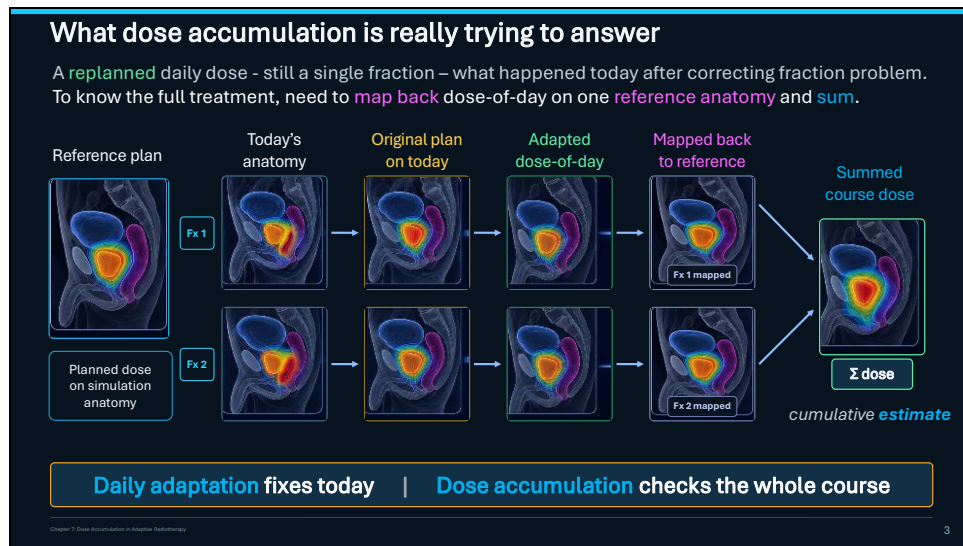
- Organs fill and empty
- Tumors can shrink, grow, or move
- Respiration changes.
- Setup geometry changes.

.. When the anatomy changes, dose received that day changes even when the machine delivers exactly what was planned.

We may miss the target; we may give more to an OAR.

... So, to address the change, we adapt.

We create a plan of the day can restore target coverage and help protect organs at risk **for that specific fraction**.



But the replanned (adapted) dose-of-day is still only a single fraction. It tells us what happened today after correcting the fraction's problem.

To know the full treatment, we need to map each daily dose back to a reference anatomy (typically the planning CT) and sum those mapped doses.

Once fraction 1 and fraction 2 are in the same reference geometry, they can be summed. If we do this for all fractions, the summed course dose is the cumulative estimate for the entire treatment.

So, daily adaptation fixes today; dose accumulation checks the whole course.

Dose accumulation has an uncertainty budget

- As soon as we start mapping dose, we also inherit uncertainty.
- Dose accumulation - useful, but not magic.
- Accumulated dose is only as reliable as the assumptions behind it.

Deformation field - No algorithm perfectly models anatomy when contrast is poor or changes are abrupt.	Modality mismatch - CT, CBCT, and MR of the day differ in resolution, distortion, calibration, and contrast. - This affects registration of the images.
Dose warping assumption - Dose does not always deform with tissue in a physically simple way. - The result is only as valid as the registration behind it.	Gradient amplification A small spatial miss near a steep gradient can become a large dosimetric miss.

How to use the estimate:

Do not ask only “What is the estimate?”
Ask “Can we trust this estimate for this endpoint?”

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As soon as we start mapping dose between anatomies, we also inherit uncertainty.

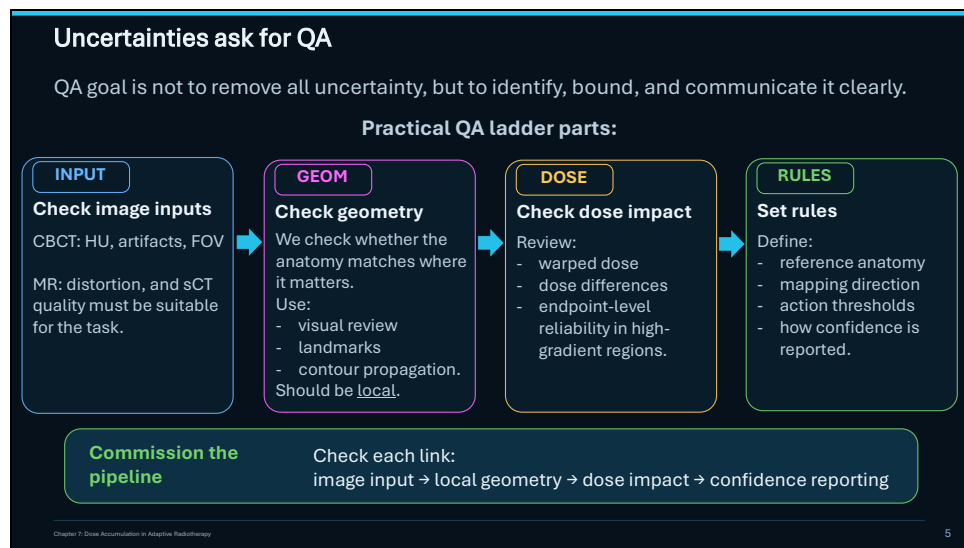
- Dose accumulation is useful, but it is not magic.
- The accumulated dose is only as reliable as the assumptions behind it.

So, what uncertainties are we facing?

- First, there is the deformation field.
... because no algorithm perfectly models anatomy when contrast is poor or changes are abrupt.
- Second, there is the dose warping assumption.
... because dose does not always deform with tissue in a physically simple way
... and the result depends on the validity of that mapping behind it.
- Third, there is modality mismatch.
... CT, CBCT, and MR of the day differ in resolution, distortion, calibration, and contrast
... which affects both density assignment and registration.
- Fourth, there is gradient amplification.
... meaning that a small spatial miss near a steep dose gradient can become a large dosimetric miss.

So we should not just ask “What is the estimate?”

The better question is “Can we trust this estimate for this endpoint?”



Uncertainties ask for QA.

The goal of the QA is not to remove all uncertainty, rather to identify it, bound it and communicate it clearly.

A practical QA ladder has four parts.

First: **input**. We check image inputs.

For CBCT, that means HU accuracy, artifacts, field of view, scatter, and truncation.

For MR, that means distortion, synthetic CT quality, and whether the image is suitable for the dose calculation task.

Second: **geometry**.

We check whether the anatomy matches where it matters.

For that, we can use visual review, landmarks, and contour propagation.

Importantly, this should be local. ***A registration may be acceptable for one structure but not another.***

Third: **dose**. We check the dose impact.

That means reviewing warped dose, dose differences, and endpoint-level reliability, especially in high-gradient regions.

Fourth: **rules**. We need to set the rules.

Meaning that we need to define the reference anatomy, the mapping direction, the action thresholds, and how confidence will be reported.

So the bottom line is: we have to commission the pipeline by checking each link: image input, local geometry, dose impact, and confidence reporting.

Audience pulse check 1

RAISE YOUR HAND

Which uncertainty is the least comfortable in your workflow?

A	B	C	D
DIR plausibility	HU / density accuracy	MR distortion or sCT fidelity	How to report uncertainty

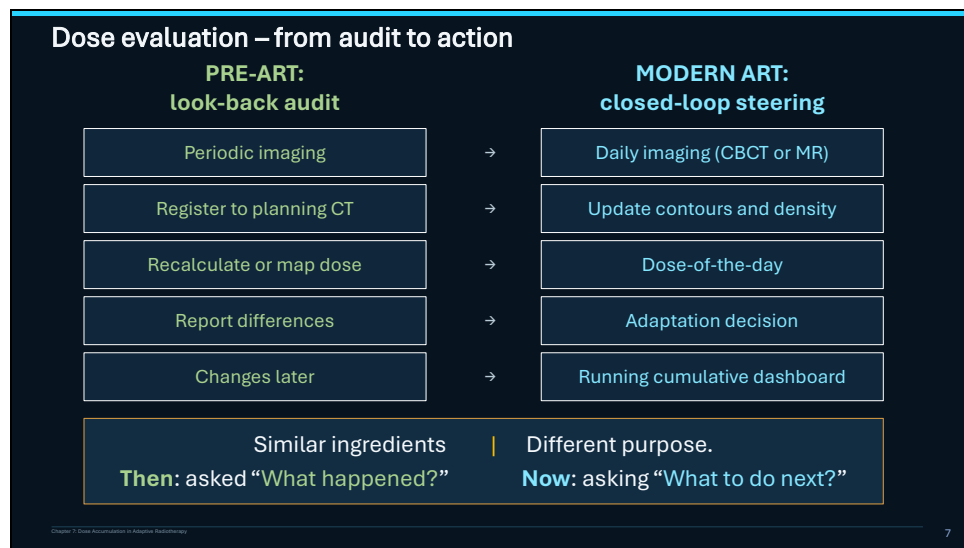
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Before moving

The reason I like this question is that there is no single universal answer. Different institutions may have different weak points depending on their imaging platform, registration tools, and clinical use case.

But the common message is the same: the goal is not to pretend uncertainty is gone. The goal is to know where it lives, how large it might be, and whether it affects the endpoint we are trying to use.



Let's review now how the daily dose evaluation evolved from audit to action.

The pre-ART workflow was a look-back audit:

- We did periodic imaging.
- The image was registered to the planning CT.
- Dose was recalculated or mapped.
- Then we reported differences from the original plan.
- The action items came later.

The modern ART workflow is a closed-loop steering:

- Imaging is daily, using CBCT or MR.
- Contours and density are updated.
- We calculate the dose-of-the-day.
- That information feeds into an adaptation decision.
- Then we run a cumulative dashboard.

So the ingredients are not completely new, but they are used with a different purpose.

We used to ask: **What happened?**

Now we ask: **What should we do next?**

THIS is the operational shift.

Audience pulse check 2

RAISE YOUR HAND

At your institution, how is cumulative dose used today in tx adaptation?

A	B	C	D
Rarely / research only	Offline review or replan support	Mid-course decision support	Online / near-real-time steering

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Another quick show of hands.

At your institution, how is cumulative dose used today in treatment adaptation?

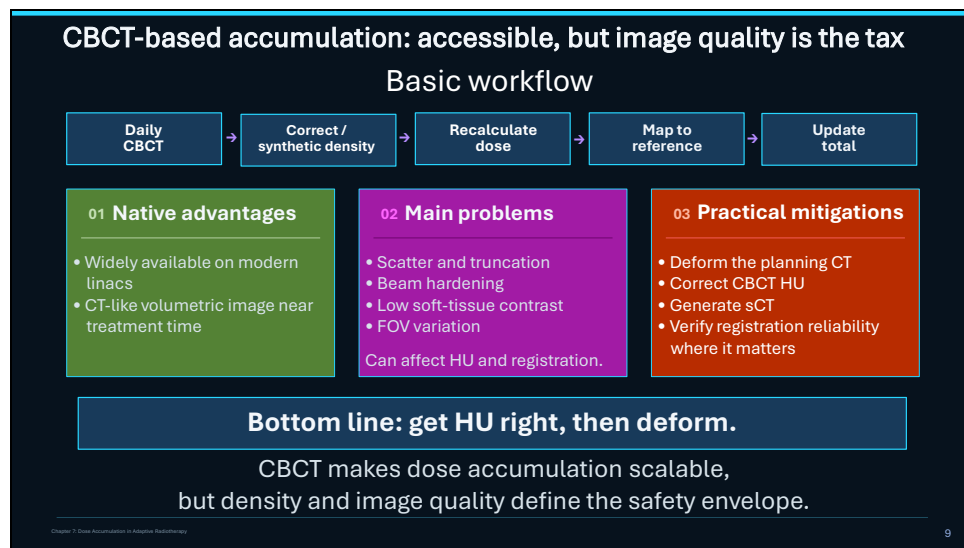
A: Rarely, or research only.

B: Offline review or replan support.

C: Mid-course decision support.

D: Online or near-real-time steering.

That spread is exactly why this topic is important. Some centers are still mainly using cumulative dose for retrospective review or selected offline decisions. Others are moving toward more prospective use.



What does dose accumulation look like when the daily image is CBCT?

The basic workflow is daily CBCT → corrected or synthetic density → recalculate dose → map to the reference anatomy → update the cumulative total.

CBCT is attractive because it is widely available on modern linacs and gives us a CT-like volumetric image near treatment time.

But the tax we pay is image quality.

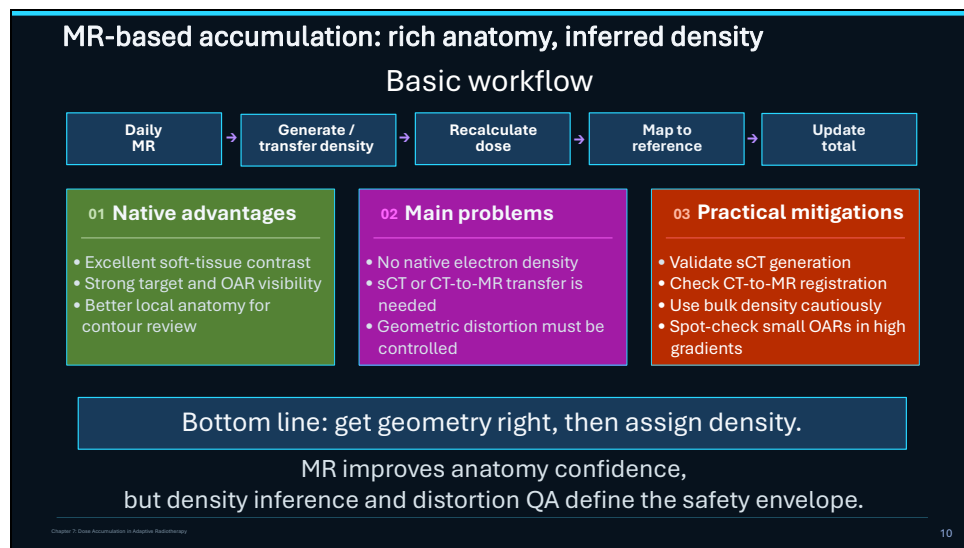
CBCT can have scatter and truncation. It can have beam hardening. It often has lower soft-tissue contrast than diagnostic CT. The field of view can vary. All these issues can affect HU accuracy and registration reliability.

That matters because dose accumulation needs both geometry and density.

The practical mitigations are to: deform the planning CT, correct CBCT HU, generate a synthetic CT, and verify registration reliability where it matters.

So the bottom line is: **get HU right, then deform.**

CBCT makes dose accumulation scalable, but density and image quality define the safety envelope.



What does dose accumulation look like when the daily image is MR?

The basic workflow is daily MR → generate or transfer density → recalculate dose → map to the reference anatomy → update the cumulative total.

MR is attractive because it gives excellent soft-tissue contrast. It improves visibility of targets and organs at risk. It can give better local anatomy for contour review.

However, MR has no native electron density, thus a sCT or CT-to-MR transfer is needed.

Moreover, the geometric distortion must be controlled.

That matters because, remember, dose accumulation needs both geometry and density.

The practical mitigations are to: Validate sCT generation, Check CT-to-MR registration, Use bulk density cautiously, Spot-check small OARs in high gradients

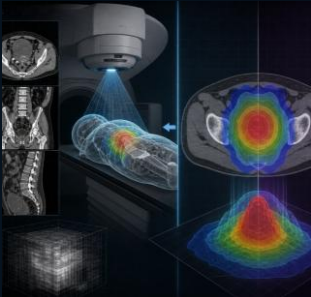
So the bottom line is: **get geometry right, then assign density.**

MR improves anatomy confidence, but density inference and distortion QA define the safety envelope.

CBCT vs MR: Are they solving the same problem?

Yes. Both help create a reliable **dose-of-the-day** that can be **mapped** to a reference anatomy and added to the **cumulative** course estimate.

But pain points are different.



CBCT
Built-in density, limited contrast
Main hurdles: scatter / HU / artifacts
QA focus: HU and anatomy reliability

MR
Excellent soft-tissue contrast, no native HU
Main hurdles: sCT / distortion
QA focus: geometry and density inference

Either way, we need:
validated deformation + meaningful dose mapping + transparent uncertainty reporting.

Modality changes the QA emphasis, but not the central responsibility.

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Now the question before us is: are CBCT and MR solving the same problem?

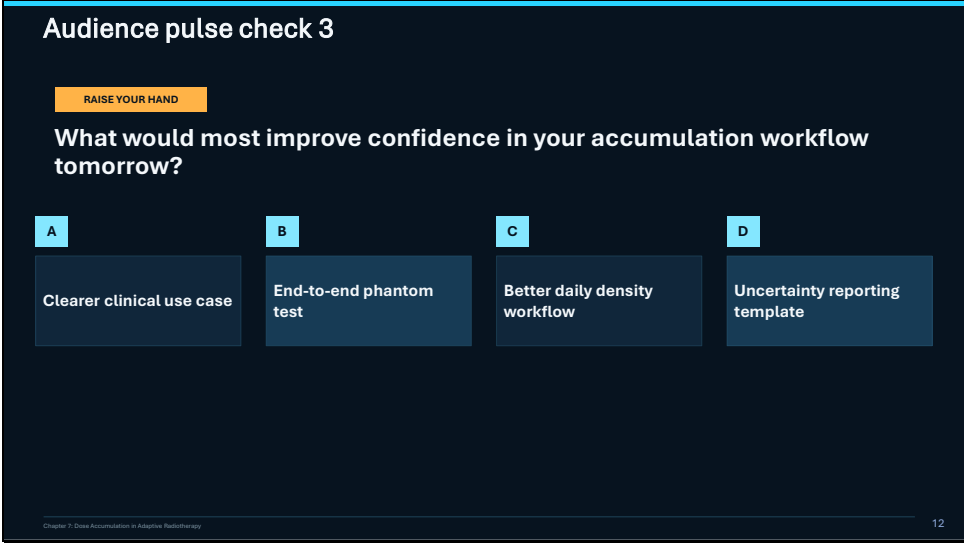
The answer is **Yes**. Both help create a reliable dose-of-the-day that can be mapped to a reference anatomy and added to the cumulative course estimate.

However, the pain points are different.

- CBCT has built-in density information, but limited contrast. Other hurdles are scatter, HU accuracy, and artifacts. So the QA focus is HU and anatomy reliability.
- MR, on the other hand, has excellent soft-tissue contrast, but no native HU. The main hurdles are synthetic CT generation and geometric distortion. So the QA focus is geometry and density inference.

Either way, we need validated deformation, meaningful dose mapping, and transparent uncertainty reporting.

So the modality changes the QA emphasis, but not the central responsibility.

The slide is titled "Audience pulse check 3" in white text on a dark blue background. Below the title is an orange button with the text "RAISE YOUR HAND". Underneath the button is the question "What would most improve confidence in your accumulation workflow tomorrow?". At the bottom of the slide, there are four options labeled A, B, C, and D, each in a light blue box. Option A is "Clearer clinical use case", Option B is "End-to-end phantom test", Option C is "Better daily density workflow", and Option D is "Uncertainty reporting template". At the very bottom, there is a small footer that reads "Chapter 7: Dose Accumulation in Adaptive Radiotherapy" on the left and the number "12" on the right.

Audience pulse check 3

RAISE YOUR HAND

What would most improve confidence in your accumulation workflow tomorrow?

A Clearer clinical use case

B End-to-end phantom test

C Better daily density workflow

D Uncertainty reporting template

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Last quick show of hands.

What would most improve confidence in your accumulation workflow tomorrow?

A: A clearer clinical use case.

B: An end-to-end phantom test.

C: A better daily density workflow.

D: An uncertainty reporting template.

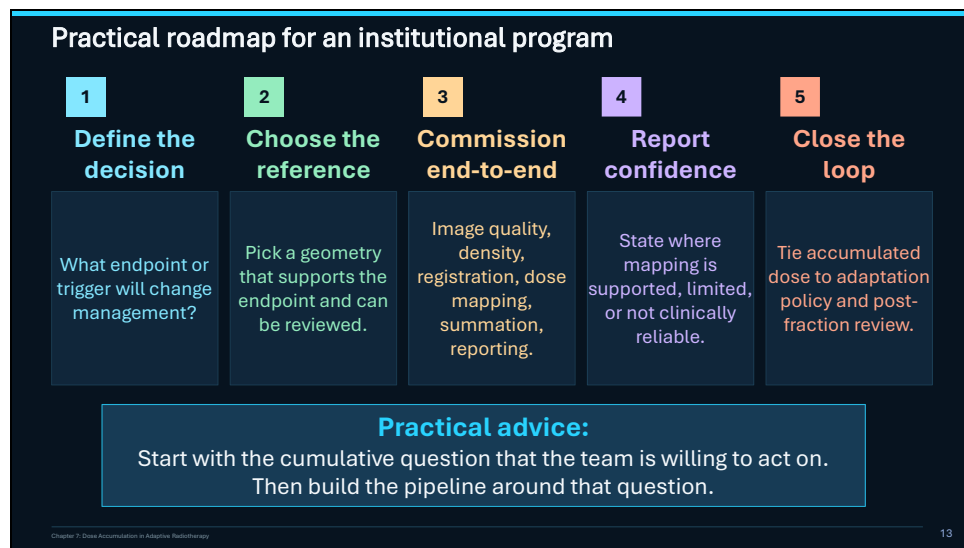
This is a very practical question because all four answers are reasonable. But they point to different problems.

If the answer is A, the issue may be that the team has the technology but not a clear decision pathway.

If the answer is B, the issue is commissioning and end-to-end validation.

If the answer is C, the bottleneck is density and daily dose calculation.

If the answer is D, the bottleneck is communication: how to report confidence so the accumulated dose can be interpreted responsibly.



Opening question: *How should an institution build dose accumulation into practice?*

- Step one is to **define the decision**. What endpoint or trigger will actually change management? If the team is not willing to act on the cumulative dose result, the workflow may not need to be built at that level of complexity.

Step two is to **choose the reference**. Pick a geometry that supports the endpoint and can be reviewed. The best reference anatomy depends on the question being asked.

Step three is to **commission end-to-end**. That means image quality, density, registration, dose mapping, summation, and reporting. The whole chain matters.

Step four is to **report confidence**. State where mapping is supported, where it is limited, and where it is not clinically reliable.

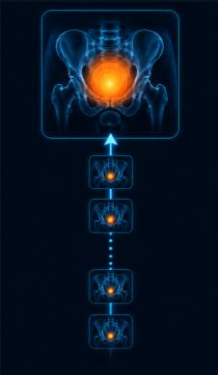
Step five is to **close the loop**. Tie accumulated dose to adaptation policy and post-fraction review.

So the practical advice is: start with the cumulative question the team is willing to act on. Then build the pipeline around that question.

Close: adaptive therapy needs cumulative dosimetry

- 01 Dose accumulation bridges plan and reality**
It translates multiple fraction-specific dose distributions into a cumulative clinical estimate.
- 02 It is only meaningful with QA and uncertainty context**
Registration, density, distortion, and mapping assumptions must be validated and reported.
- 03 The future is decision-linked accumulation**
Cumulative dose becomes a trigger for adaptation, toxicity prediction, and course-level steering.

**We treat fractions,
but we judge the total.**



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Why does adaptive therapy need adaptive dosimetry?

- Dose accumulation bridges plan and reality. It translates multiple fraction-specific dose distributions into a cumulative clinical estimate.
 - But it is only meaningful with QA and uncertainty context. Registration, density, distortion, and dose mapping assumptions must be validated and reported.
 - And the future is decision-linked accumulation. Cumulative dose can become a trigger for adaptation, toxicity prediction, and course-level steering.
- So the closing message is the same as the opening message: **we treat fractions, but we judge the total.**

Thank you. I would be happy to take questions or hear how people are currently using accumulated dose in their own adaptive workflows.